📘 Game Theory DP – Winning & Losing States (Clean Notes)

(Based on your code and logic)

🎯 Problem Concept

Ek game hai jisme player ek number `n` se start karta hai.

Players alternate moves karte hain.

Allowed moves:

- 1 hatana
- `k` hatana
- `l` hatana

Jis player ke paas move na bache → **wo game haar jata hai**.

🔥 Goal

Har `n` ke liye decide karna:

- **A** wins? → print `"A"`
- **B** wins? → print `"B"`

🎮 Game Theory Basics

Game states do types ke hote hain:

✔️ 1. **Winning State (W)**

Agar current player koi aisa move kar sakta ho jo opponent ko **losing state** me daal de.

❌ 2. **Losing State (L)**

Agar **saare possible moves opponent ko winning state** me hi daalte ho → tum haaroge.

🧠 DP Approach

`dp[i] = true`

→ state `i` ek **winning state** hai.

`dp[i] = false`

→ state `i` ek **losing state** hai.

📌 Base Cases

```
dp[1] = true;   // 1 rakhe hain → A can take 1 → A wins
dp[k] = true;   // exactly k stones → A takes k → wins
dp[l] = true;   // exactly l stones → A takes l → wins
```

🧮 Transition

A loses only if:

- dp[i–1] = W (winning for opponent)
- dp[i–k] = W (winning for opponent)
- dp[i–l] = W (winning for opponent)

Yani:

```
dp[i] = !( dp[i-1] && dp[i-k] && dp[i-l] );
```

But agar move possible nahi hai (`i-k < 1`), tumne usko default `true` treat kiya hai:

```
(i-k >= 1) ? dp[i-k] : 1
```

Yani tum keh rahe ho:

> Invalid move = "opponent ke liye winning" treat karo
>
> jisse wo move effectively ignore ho jata hai.

This is a fine trick and works for this problem.

🟦 Final Logic Summary

Condition	Meaning
`dp[i] = true`	A ke liye winning
`dp[i] = false`	B ke liye winning

Transition:

A wins at state `i` if **Koi bhi ek** move opponent ko losing banaye.

A loses if **saare** moves opponent ko winning banaye.

🎯 Time & Space Complexity

- **DP Size:** 1,000,000
- **Time:** `O(N)`
- **Space:** `O(N)`

🧾 Output Explanation

For every query `n`:

```
dp[n] = true  → print A
dp[n] = false → print B
```

⭐ Final Points (important for exams/interviews)

- Ye ek **Classic impartial game** hai (players identical moves).
- Ye **DP + Game Theory** ka example hai.
- Winning/Losing states: **“If any move leads to a losing state → winning state”**
- Condition `"!(...)"` logically represents losing state check.
- Tumhara formula **correctly optimized** hai.